

Economics 22 - Macroeconomic Variables updated

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Dartmouth College

- GDP
 - Definition
 - Components
 - Real vs. nominal
- Prices
 - GDP Deflator
 - CPI
- Unemployment

Basic US Economic Numbers

- GDP = \$30.8 Trillion (2025, annual, nominal)
- US Population = 343 million (January 2026, est.)
- World Population = 8.2 billion (January 2026, est.)
- Unemployment Rate = 4.3 (Apr 2026)
- Inflation = 3.8 (CPI-all items, Apr 2025 - Apr 2026)
- Core inflation = 2.7 (CPI-all items less food and energy, Apr 2025 - Apr 2026)
 - Sources: <http://www.census.gov/popclock/>, <http://www.bea.gov/>,
<http://www.bls.gov/>

Why it is good to know GDP



Alexandria Ocasio-Cortez



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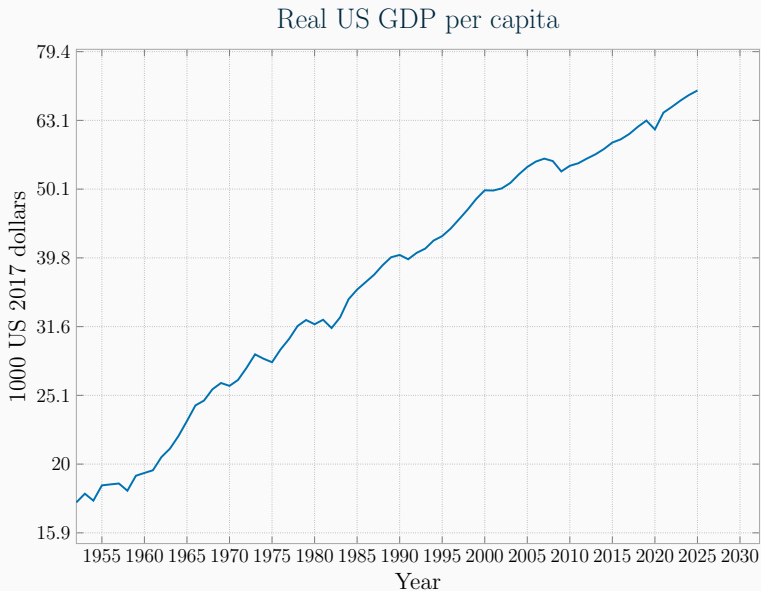


\$21 TRILLION of Pentagon financial transactions “could not be traced, documented, or explained.”

\$21T in Pentagon accounting errors. Medicare for All costs ~\$32T.

That means 66% of Medicare for All could have been funded already by the Pentagon.

And that's before our premiums.



Measuring GDP

- Lots of things go into GDP
 - Millions of Chickens
 - Billions of paperclips
 - Thousands of econ lectures
- How do we add Apples and Oranges?

- The natural metric for adding GDP is money prices.
- Prices provide the relative value of things in an economy.
- This turns out to be one of money's most important functions.

Measuring GDP

- Suppose that an economy had only two goods.
 - Apples
 - Oranges
- GDP in terms of prices would be

$$GDP = (Q_A \times P_A) + (Q_O \times P_O)$$

- One way of measuring GDP would be to measure the price and quantity of everything sold in the country.
- There is a second way – we take advantage of circular flows in the economy.

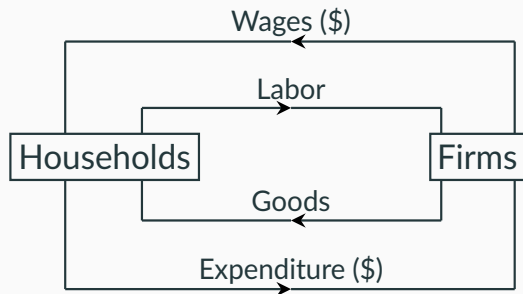
Measuring GDP

- Imagine an economy with only one good, bread, produced using only labor.

There are two types of actors in this economy:

- Households
 - Provide labor to make bread
 - Get paid wages by the firm (which they use to buy bread)
- Firms
 - Pay workers to make bread
 - Sell bread to households

Circular Flow Diagram (simple)



- Households earn wages from firms. All the income is in the form of wages.
- Households spend all their income on goods.
- GDP is both therefore both national income and national expenditure.

Complications in Counting GDP

- Used Goods
 - How do you count the sale of a used car?
- Intermediate Goods
 - Firm 1 makes Flour – sells it to Firm 2
 - Firm 2 makes bread – sells it to consumers
 - How do we count up GDP?

- Adding up the value of flour and bread would be double counting.
- We can fix this in two ways
 - Just count final goods – in this case bread.
 - Add up the value added at each step; the value of the output minus the value of the inputs.

Complications

- Non-market goods
 - Home cooked meals
 - Child care
 - Snow Shoveling
- Imputed value
 - Housing services
 - Government services

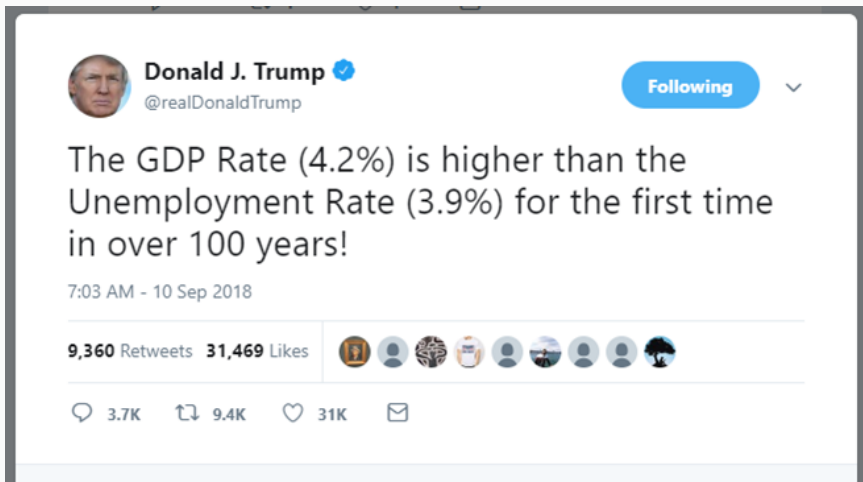
Stocks vs. Flows

- Think of a bathtub
 - Stock – level of water in the tub at any time
 - Flow – quantity of water flowing in/out of the tub
- Goods aren't produced instantly -- they are produced over time.
 - We don't say a factory makes 100 loaves of bread, we say they make 100 loaves per day.

Stocks vs. Flows

- Stock – quantity measured at any point in time.
 - Credit Card Balance
 - Capital stock
 - Number of unemployed people
- Flow – Quantity measured per unit of time.
 - Credit card charges/payments
 - Investment
 - Number of people losing their jobs

Stocks vs flows == rates vs levels



- There are four uses to which GDP can be put

$$Y = C + I + G + NX$$

- This is called the national income accounting identity.

$$Y = C + I + G + NX$$

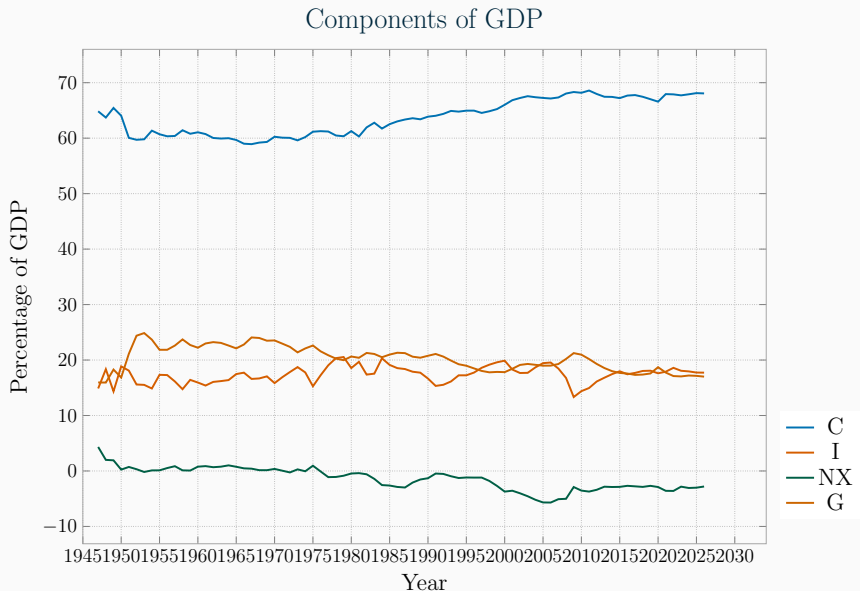
- C – Consumption of goods and services by households.
- I – Investment. Investment is anytime you buy something for use in the future. Houses, factories.
- G – Government spending on goods and services – like tanks and teachers. Note that this does not include transfer payments.
- NX – Net exports. This is the amount that we export, less the amount that we import.

How large are the components?

All values are annual nominal values for 2025

Y	\$30.8T	Percent of GDP
C	\$21.0T	68%
I	\$5.5T	18%
G	\$5.3T	17%
NX	-\$926B	-3%

Components over time - Consumption



- Q: Given our framework for calculating GDP, what would happen if prices doubled?
- A: Nominal GDP would double
- Would this mean anything real?

- Since we want a measure of real stuff produced we need to filter out changes in prices.
- We do this by using a uniform set of prices.

Constructing Real GDP

- Find the price of each good in a given base year.
- Take the quantity of each good produced during the current year and multiply it by the price of that good in the base year.
- Sum the value of all the goods produced.
- Because the only thing changing over time is the quantity of goods, this gives us the changes in the quantity of stuff, irrespective of changes in prices.

- Suppose we want to construct RGDP in 2026 using 2017 as the base year. In our economy with apples and oranges:

$$RGDP_{2026} = (Q_{A,26} \times P_{A,17}) + (Q_{O,26} \times P_{O,17})$$

- This gives us RGDP for 2026 in 2017 prices

- How would you calculate RGDP for 2017 using 2017 as the base year?

Some numbers

- Nominal GDP in 2025– \$30.8 T
- Nominal GDP in 2017– \$19.6 T
- Real GDP in 2025– \$23.9 T (in 2017\$)
- Real GDP in 2017– \$19.6 T (in 2017\$)

- The nominal increase was \$11.2 T
- The real increase was \$4.3 T
- The increase due to inflation was \$6.9 T

- Suppose that now we want to know how much prices have gone up.
- There are two major price indices that we use. One index is called the GDP deflator (also called the “implicit price deflator for GDP”).

$$GDPD = NGDP / RGDP$$

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- For example, for 2025:

$$GDPD = \frac{(Q_{A,25} \times P_{A,25}) + (Q_{O,25} \times P_{O,25})}{(Q_{A,25} \times P_{A,17}) + (Q_{O,25} \times P_{O,17})}$$

- Note that GDPD is 1 for the base year

The GDP deflator - example

- $NGDP_{25} = \$30.8 \text{ T}$
- $RGDP_{25} = \$23.9 \text{ T (2017\$)}$
- $GDPD = NGDP / RGDP = 1.290$
- Prices rose by about 29.0% between 2017 and 2025

- The GDPD measures the prices of all the goods and services that are produced in the country.
- What if we wanted to measure the price of a different bundle?
 - Cost of being a Dartmouth student
 - Cost of being caffeinated for your 9am classes.

Caffeine Price Index

- Suppose you got data on the prices of coffee and tea.
- Q: How much has the price of caffeine risen?

	2025	2026
Tea	1.00	1.20
Coffee	1.00	1.40

Caffeine Price Index

- A: You can't tell because you don't know how much to weight the two components.
- We know that tea has gone up by 20% and that coffee has gone up by 40%.
- Overall, the price of caffeinated beverages has gone up by somewhere between the two. The overall change will be a weighted average. But what weights to use?

- Suppose that we did a survey and found that in 2025 (the base year), consumption of the two beverages was:
 - Tea – 1000 cups
 - Coffee – 3000 cups
- We can use these quantities as weights in our price index. Coffee will have three times the weight of tea.

- Our caffeine consumption cost is:
 - $1000 * 1.0 + 3000 * 1.0 = \$4,000$ in 2025
 - $1000 * 1.2 + 3000 * 1.4 = \$5,400$ in 2026
- The price of caffeine had increased by 35%.

PNC Cost of Christmas

PNC Christmas Price Index - 2008

	Traditional			Internet		
	<u>2007</u>	<u>2008</u>	<u>% Change</u> <u>2008/07</u>	<u>2007</u>	<u>2008</u>	<u>% Change</u> <u>2008/07</u>
One Partridge in a Pear Tree	\$ 164.99	\$ 219.99	33.3%	\$ 224.94	\$ 235.00	4.5%
Partridge	\$ 15.00	\$ 20.00	33.3%	75.00	85.00	13.3%
Pear Tree	\$ 149.99	\$ 199.99	33.3%	149.94	150.00	0.0%
Two Turtle Doves	\$ 40.00	\$ 55.00	37.5%	130.00	130.00	0.0%
Three French Hen	\$ 45.00	\$ 30.00	-33.3%	195.00	195.00	0.0%
Four Calling Birds	\$ 599.96	\$ 599.96	0.0%	455.00	475.00	4.4%
Five Gold Rings	\$ 395.00	\$ 349.95	-11.4%	329.75	329.75	0.0%
Six Geese-a-Laying	\$ 360.00	\$ 240.00	-33.3%	660.00	660.00	0.0%
Seven Swans-a-Swimming	\$ 4,200.00	\$ 5,600.00	33.3%	7,035.00	7,035.00	0.0%
Eight Maids-a-Milking	\$ 46.80	\$ 52.40	12.0%	310.08	310.08	0.0%
Nine Ladies Dancing	\$ 4,759.19	\$ 4,759.19	0.0%	7,106.00	7,237.00	1.8%
10 Lords-a-Leaping	\$ 4,285.06	\$ 4,413.61	3.0%	11,283.23	11,629.79	3.1%
11 Pipers Piping	\$ 2,213.40	\$ 2,284.80	3.2%	1,900.00	2,100.00	10.5%
12 Drummers Drumming	\$ 2,397.85	\$ 2,475.20	3.2%	1,620.00	1,620.00	0.0%
Total Christmas Price Index	\$ 19,507.25	\$ 21,080.10	8.1%	\$ 31,249.00	\$ 31,956.62	2.3%
% change						
True cost of Christmas in song	\$ 78,100.10	\$ 86,608.51	10.9%	\$ 128,886.37	\$ 131,150.76	1.8%
% change						
"Core" index, excluding swans	\$ 15,307.25	\$ 15,480.10	1.1%	\$ 24,214.00	\$ 24,921.62	2.9%

- To calculate the CPI, you choose a base year and a basket of goods.
- The CPI compares the cost of the basket today versus the cost in the base year.

$$CPI_t = 100 \times \frac{\sum p_t \times q_{base}}{\sum p_{base} \times q_{base}}$$

- The basket of goods for the US is chosen by the government.
- The government performs a survey to find out what people consume and in what proportions. This determines the weights.

- Suppose that you wanted to compensate people for the increased cost of their caffeine consumption.
 - This is the idea behind cost of living adjustments (COLAs).
- What if people only slightly prefer coffee to tea?

- If people switch from coffee to tea, the increase in the price index overstates the needed increase in people's caffeine budgets.
- If you increased everyone's caffeine budget by 35% (the amount that the caffeine CPI has risen), they would be better off than they were before, because they are willing to switch to tea.

- The problem with indexes based on a fixed basket of goods is it does not take account of the fact that people will switch from expensive products to cheaper ones.
- This is an unavoidable problem with this sort of index.

Differences between CPI and GDPD

- CPI

- Basket of goods is just consumer goods
- Goods produced abroad are counted
- The basket of goods is from the base year (Laspeyres index, fixed basket)

- GDPD

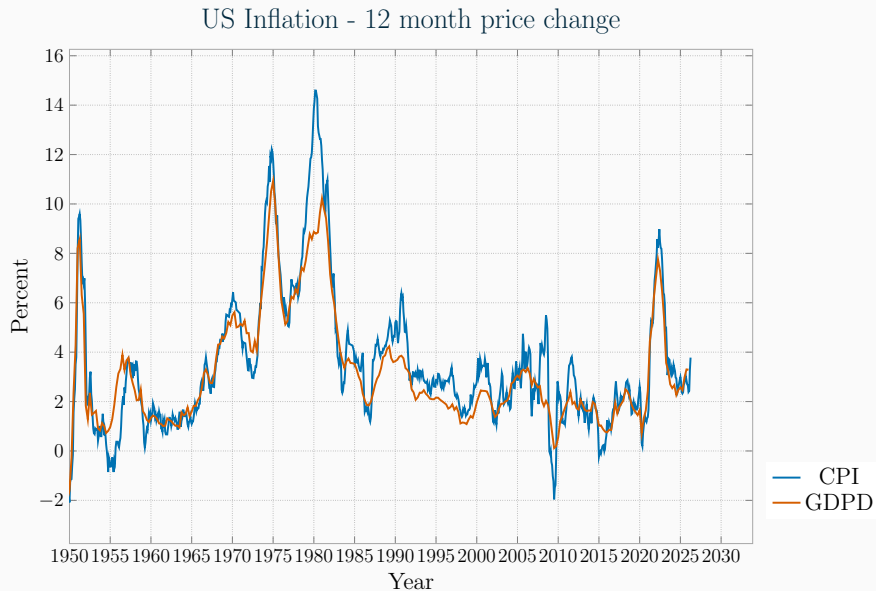
- Basket of goods is all goods
- Goods produced abroad are not counted
- The basket of goods is from the current year (Paasche index, changing basket)

Problems with choosing baskets

- Suppose that we use a base year consumption basket from 50 years ago
 - the typical basket that the average person got with the average salary.
- Can we consume it now with an average salary?
- Maybe, but some things have gone up tremendously in value, like land.

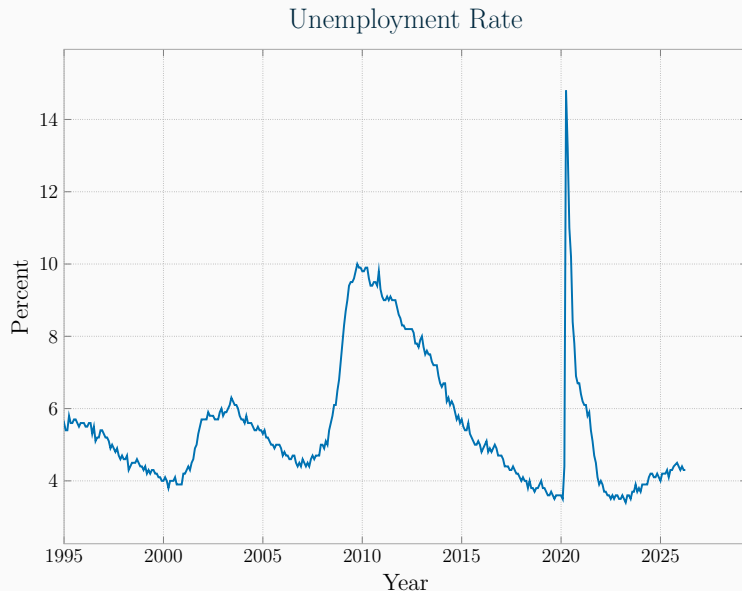
Problems with choosing baskets

- On the other hand, suppose that we use today's average consumption basket.
- Could people in the past have consumed it with the average income?
- No. we have things like iPhones, HDTV's and computers that they didn't have then, and other things are much cheaper now, like transatlantic transportation.



- Perhaps the most watched economic statistic
- Why?
 - A well functioning economy uses all resources. Unemployment wastes resources.
 - The costs are unevenly distributed

Unemployment



- To figure out unemployment, the government does a survey each month.
- They divide the population into three groups.

$$\textit{Adult population} = (E + U) + \textit{OLF}$$

- E – employed
- U – unemployed
- OLF – out of labor force (students, retired, stay at home parents, etc.)

Unemployment

- The labor force consists of the Employed and the Unemployed

$$LF = E + U$$

- We can then form measures:
 - Unemployment Rate

$$UR = U/LF$$

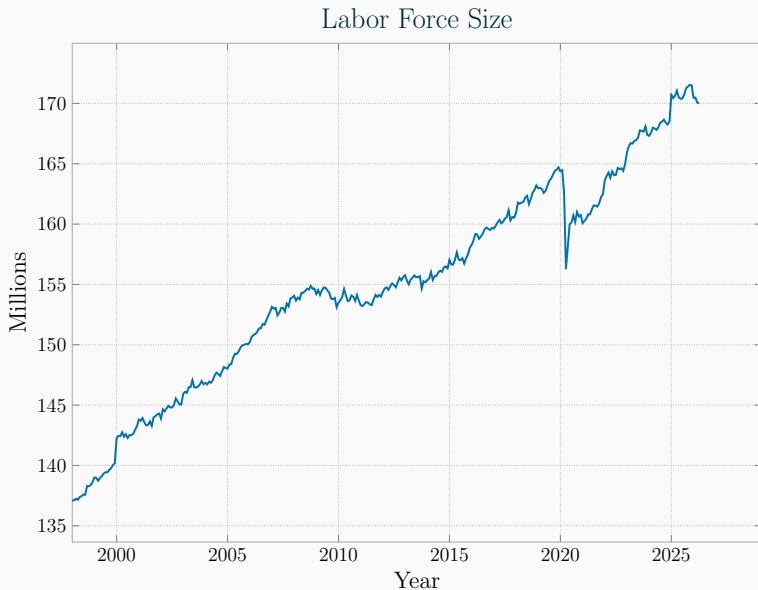
- Labor force Participation Rate

$$LFPR = LF / adult\ pop$$

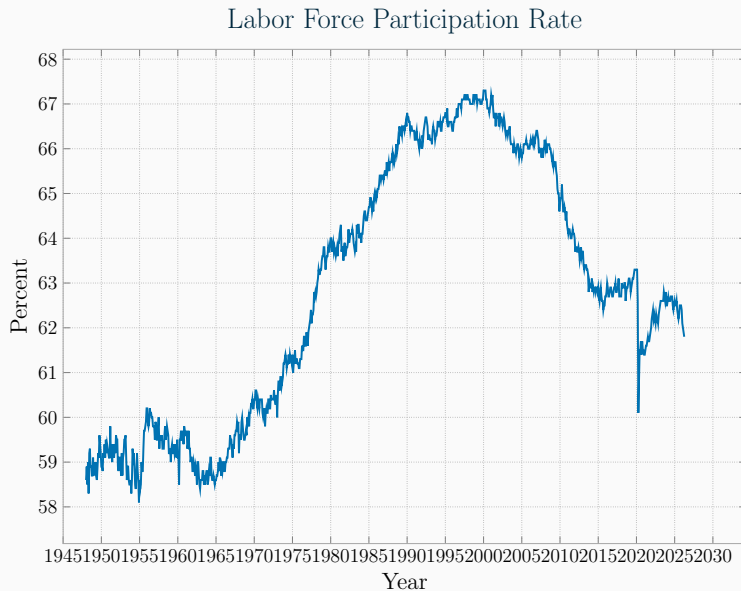
- Employment to Population Ratio

$$EPOP = E / adult\ pop$$

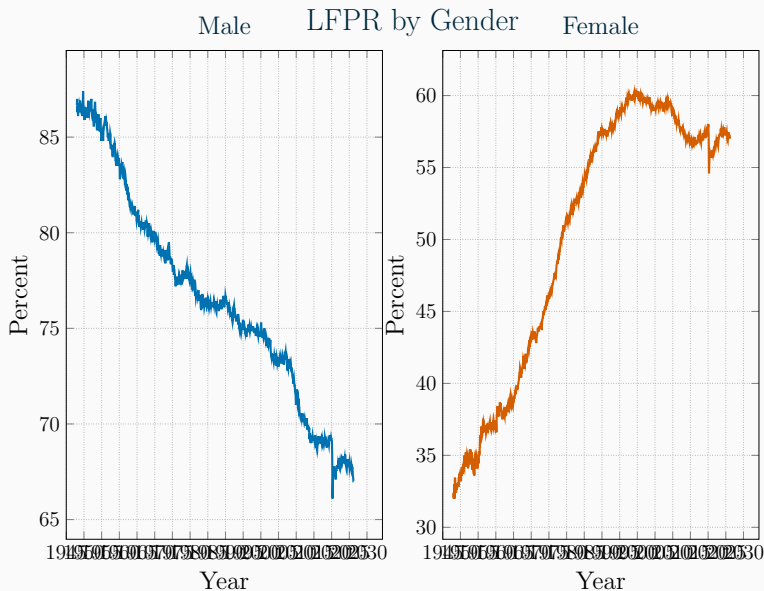
Labor Force Size



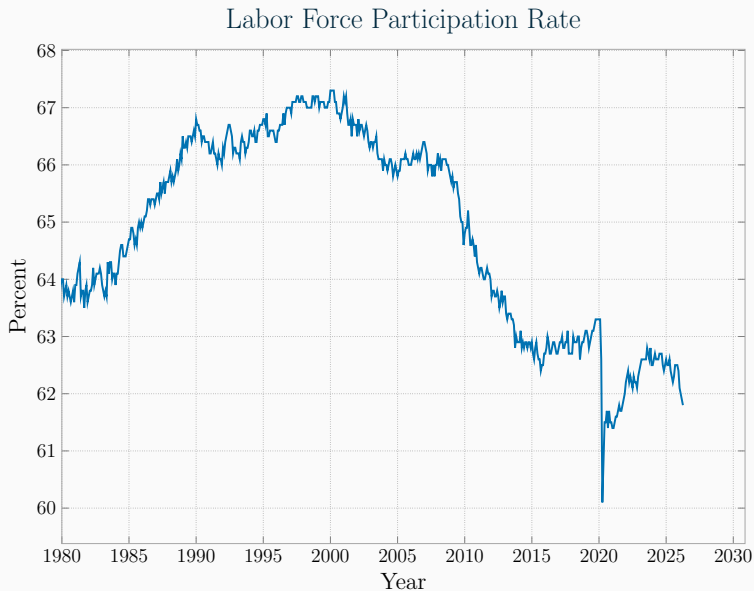
Labor Force Participation Rate



LFPR by Gender

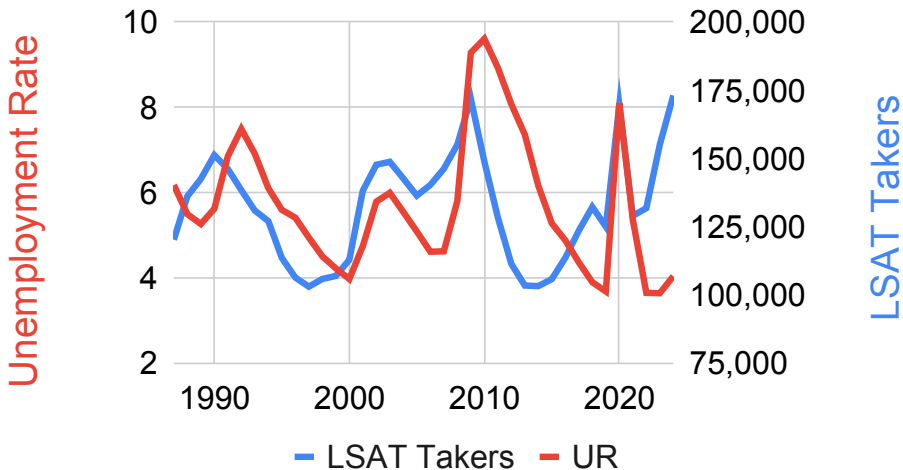


Labor Force Participation Over the Business Cycle

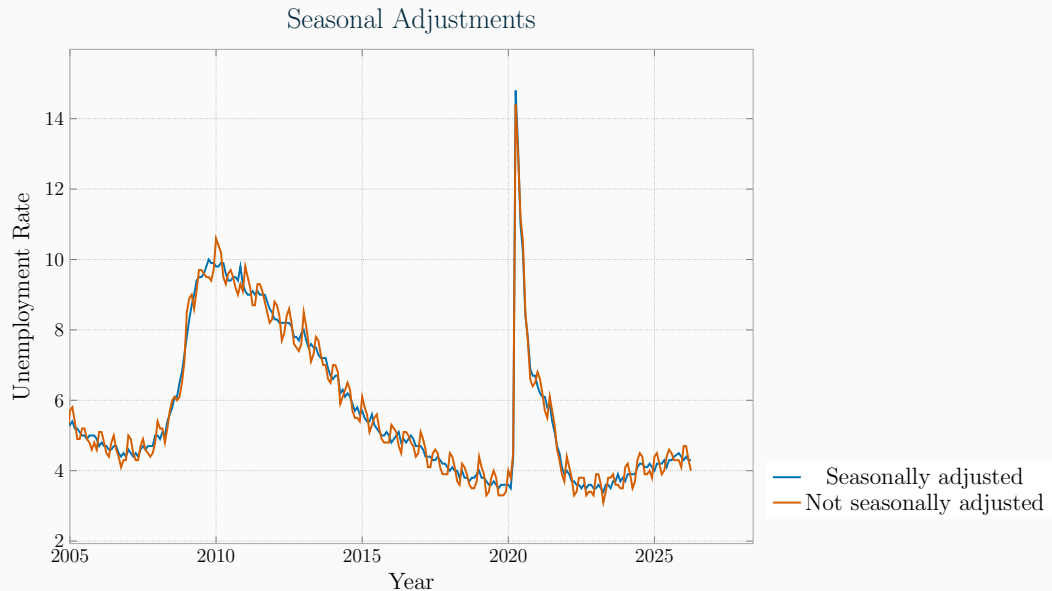


LSAT tests taken vs. Unemployment

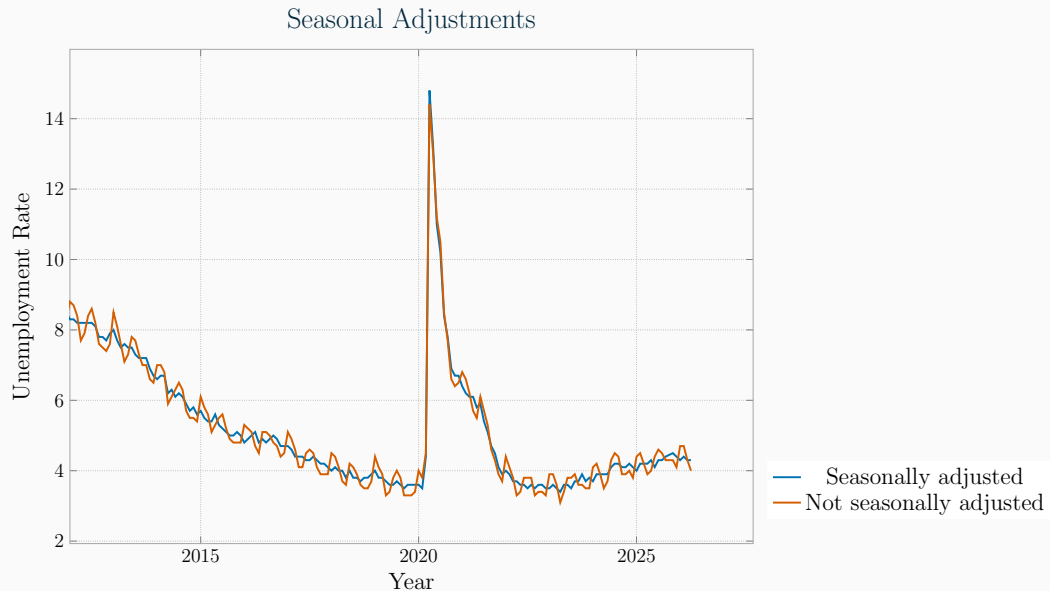
LSAT takers Versus the Unemployment Rate



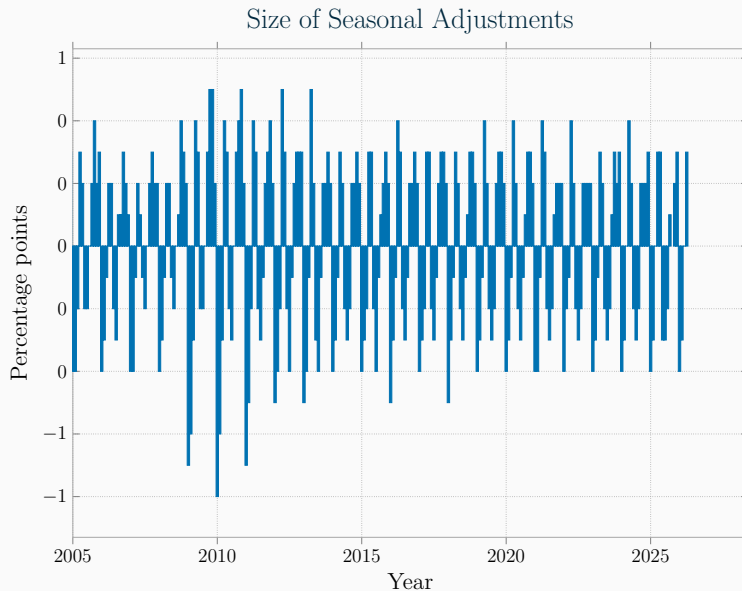
Seasonal Adjustments



Seasonal Adjustments



Size of Adjustments



- When unemployment is high, GDP growth tends to be low.

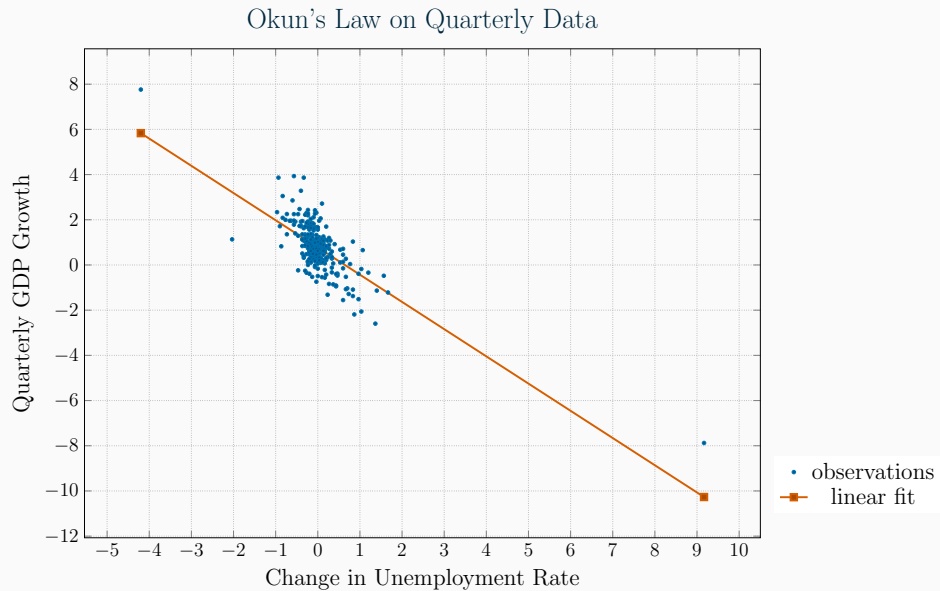
$$\text{Real GDP Growth} = 3\% - 2 * \Delta \text{Unemployment Rate}$$

- If the UR isn't changing, Real GDP growth is about 3%.
- For each percentage point increase in unemployment, real GDP growth declines by about 2

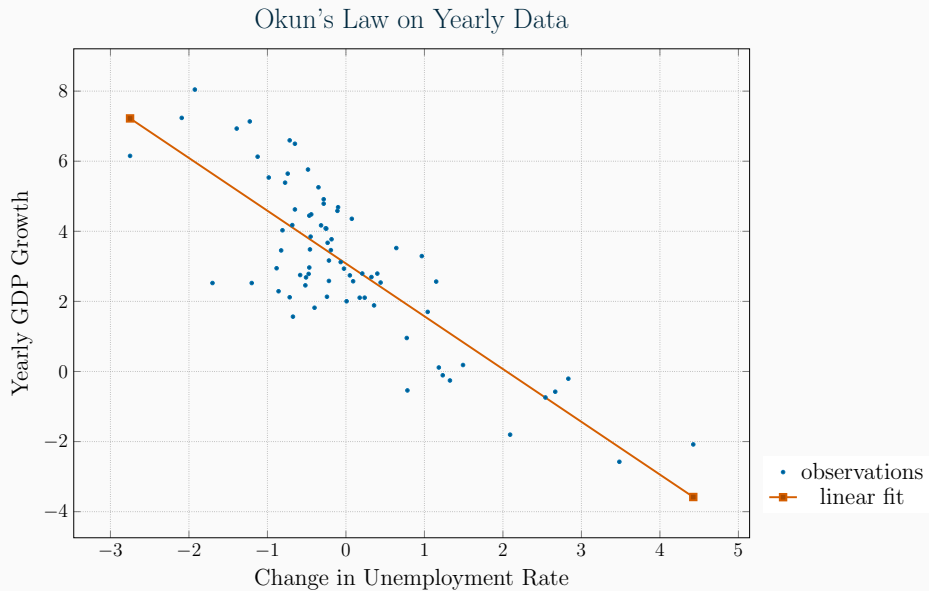
$$\text{Real GDP Growth} = 3\% - 2 * \Delta \text{Unemployment Rate}$$

- To reduce unemployment by two percentage points the US needs to grow at about 7%

Okun's Law on Quarterly Data



Okun's Law on Yearly Data



Okun's Law Regression Results

VARIABLES	(1) Yearly	(2) Quarterly
Change in Unemployment Rate	-1.50*** (0.13)	-1.21*** (0.06)
Constant	3.09*** (0.16)	0.78*** (0.04)
Observations	74	310
R-squared	0.65	0.56